

MATSU NANGAN, Taiwan

WMO: 466890

Lat: 26.160N

Lon: 119.958E

Elev: 223

StdP: 14.58

Time Zone: 8.00 (E08)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	40.1	42.1	31.5	26.0	45.3	33.7	28.5	45.4	34.6	43.8	32.0	44.5	19.2	30	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	88.9	81.9	87.6	81.6	86.1	80.8	82.8	87.0	81.9	85.9	81.0	84.8	6.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	166.3	86.5	80.7	160.8	85.6	79.9	156.4	84.5	47.2	87.6	46.1	86.6	45.1	85.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.0	30.3	26.9	36.9	91.5	3.0	2.7	34.7	93.5	33.0	95.1	31.3	96.6	29.2	98.6
			34.3	85.1	1.6	0.9	33.2	85.8	32.2	86.3	31.3	86.8	30.1	87.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	50.1	49.8	53.0	60.4	69.4	76.6	81.4	81.5	78.0	70.7	63.9	55.9	(6)
	DBStd	12.27	4.89	4.75	4.76	5.78	4.75	4.46	2.26	2.12	3.33	4.05	4.78	4.43	(7)
	HDD50	148	61	56	22	2	0	0	0	0	0	0	0	7	(8)
	HDD65	1795	462	425	372	158	14	0	0	0	0	5	76	283	(9)
	CDD50	5980	63	51	116	315	602	798	973	975	841	641	416	189	(10)
	CDD65	2155	0	0	1	21	152	348	508	510	391	182	41	1	(11)
	CDH74	16716	0	0	1	21	425	2495	5192	5280	2939	348	14	1	(12)
	CDH80	3976	0	0	0	1	44	523	1431	1440	523	13	1	0	(13)
(14) Wind	WSAvg	13.3	16.2	15.3	13.9	11.1	10.5	10.3	9.7	10.6	12.5	16.3	16.9	16.3	(14)
(15) Precipitation	PrecAvg	68.1	2.8	3.8	5.7	5.9	8.4	10.2	7.0	10.2	6.5	2.6	2.7	2.3	(15)
	PrecMax	94.8	8.6	10.0	12.5	13.4	15.4	17.5	16.9	19.6	22.6	7.5	8.9	7.6	(16)
	PrecMin	36.9	0.4	0.7	2.1	2.0	1.8	3.1	0.8	2.6	1.0	0.1	0.2	0.2	(17)
	PrecStd	13.2	2.0	2.2	2.4	2.4	3.3	3.3	4.0	5.1	4.9	2.2	2.0	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	66.0	69.9	76.7	84.2	89.5	90.1	89.9	89.2	81.2	75.6	69.8	(19)
		MCWB	60.1	61.6	65.6	72.4	77.9	82.3	82.6	82.0	82.0	74.4	69.9	64.9	(20)
	2%	DB	61.2	62.3	65.9	73.5	80.5	86.4	89.0	88.4	86.4	79.1	73.3	65.9	(21)
		MCWB	56.2	59.2	62.6	70.0	75.1	80.9	82.1	81.7	81.0	73.3	68.9	62.3	(22)
	5%	DB	59.0	59.3	63.0	71.3	78.0	84.6	87.6	87.5	84.5	77.3	71.6	64.1	(23)
		MCWB	54.9	56.6	60.4	68.1	73.7	80.0	81.6	81.5	79.4	72.5	67.8	60.8	(24)
	10%	DB	57.0	56.8	60.6	68.9	76.1	82.8	85.9	86.0	82.7	75.5	69.9	62.2	(25)
		MCWB	53.4	54.0	58.1	66.2	72.5	78.8	80.7	80.7	78.2	71.0	66.3	59.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.8	65.9	72.8	78.7	83.4	84.5	84.2	83.3	76.9	71.7	65.5	(27)
		MCDB	64.7	65.0	69.0	75.4	83.0	87.6	89.1	88.1	87.2	79.6	73.4	68.4	(28)
	2%	WB	57.2	59.5	63.4	70.6	76.3	81.6	82.7	82.7	81.5	75.3	69.9	63.2	(29)
		MCDB	60.5	61.8	65.5	73.2	79.4	85.4	87.0	86.8	85.2	78.0	72.2	65.1	(30)
	5%	WB	55.4	57.0	60.9	68.5	74.6	80.1	81.8	81.8	80.0	73.6	68.5	61.2	(31)
		MCDB	58.4	59.5	62.8	70.9	77.4	83.6	85.8	85.8	83.2	76.5	70.9	63.2	(32)
	10%	WB	53.7	54.6	58.4	66.5	73.2	79.0	80.8	80.8	79.0	71.8	66.9	59.1	(33)
		MCDB	56.7	56.4	60.3	68.7	75.6	82.0	84.7	84.8	81.9	75.1	69.6	61.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.0	8.6	9.6	9.6	10.1	9.7	9.1	8.0	7.1	7.5	7.3	(35)
		MCDBR	12.2	14.2	14.0	13.2	12.5	11.8	12.0	11.4	10.5	9.0	9.5	11.1	(36)
	5% WB	MCWBR	9.4	11.2	11.2	10.2	9.4	8.6	7.8	7.4	6.9	6.4	6.9	8.7	(37)
		MCWBR	11.1	13.0	12.9	13.1	11.9	11.1	10.4	10.2	9.7	8.3	8.7	10.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.580	0.633	0.708	0.707	0.623	0.572	0.541	0.571	0.564	0.571	0.562	0.555	(40)
		taud	1.758	1.685	1.572	1.628	1.846	1.998	2.088	1.993	1.983	1.888	1.867	1.831	(41)
	Edn at Noon		207	208	202	208	226	237	244	236	234	222	211	207	(42)
		Edh at Noon	63	73	86	83	67	57	52	57	56	59	56	56	(43)
(44) All-Sky Solar Radiation	RadAvg		702	807	1041	1183	1330	1461	1926	1730	1432	1159	797	721	(44)
	RadStd		163	172	119	156	216	161	153	144	140	190	123	134	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	+139

Nomenclature: See separate page